Project-2: Extending the GPU-based VAMANA Approximate Nearest Neighbour Search (ANNS) graph construction implementation to handle streaming inputs (dynamic inserts and deletes)

Graph data structure is widely used as the underlying index for performing Approximate Nearest Neighbour Search (ANNS) as it provides high recalls and throughput. Also, it is scalable to handle large datasets. The VAMANA [1] is a popular graph construction technique. ANNS is computationally intensive, involving several iterations of neighbour fetching from the graph index and distance computations between the retrieved neighbours and query points.

GPUs offer massive parallelism and hence are an obvious choice for performing ANNS. We have developed a GPU-based implementation of the popular VAMANA algorithm. The source code is available here [2].

There are two phases in ANNS. Using the base dataset, the graph index is constructed offline. Subsequently, the search queries are processed online. For real-world applications, the dataset is constantly changing. For example, in an e-commerce portal, items are continually added or removed, and on YouTube, hundreds of videos are added or removed every second. Thus, the graph index must handle streaming data or dynamic updates. Such applications require the graph index to accept operations like **Insert and delete, apart from search**. FreshDiskANN [3] proposed a scheme for the VAMANA graphs to handle dynamic insertions and deletions.

This project aims to develop a GPU-based implementation of FreshDiskANN. Today, the ANNS public datasets only provide search queries. Additionally, a framework to generate the required workloads must also be designed to test the ANNS implementation with streaming updates.

References:

[1] Suhas Jayaram Subramanya, Fnu Devvrit, Rohan Kadekodi, Ravishankar Krishnawamy, and Harsha Vardhan Simhadri. 2019. DiskANN:Fast Accurate Billion-point Nearest Neighbor Search on a Single Node. In Advances in Neural Information Processing Systems32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, 8-14 December 2019, Vancouver, BC, Canada, Hanna M. Wallach, Hugo Larochelle, Alina Beygelzimer, Florenced'Alché-Buc, Emily B. Fox, and Roman Garnett (Eds.). 13748–13758. http://papers.nips.cc/paper/9527-rand-nsg-fast-accuratebillion-point-nearest-neighbor-search-on-a-single-node

[2] VAMANA Graph construction on GPU, https://github.com/jyothivedurada/BANG-Variants/tree/vamana-gpu

[3] A. Singh, S. J. Subramanya, R. Krishnaswamy and H. V. Simhadri, "Freshdiskann: A fast and accurate graph-based ann index for streaming similarity search," *arXiv preprint arXiv:2105.09613,* 2021.